

StatisticalVariable

A Schema.org Type

This term is in the "new" area - implementation feedback and adoption from applications and websites can help improve our definitions.

Thing > **Intangible** > **ConstraintNode** > **StatisticalVariable**

[more...]

StatisticalVariable represents any type of statistical metric that can be measured at a place and time. The usage pattern for **StatisticalVariable** is typically expressed using **Observation** with an explicit **populationType**, which is a type, typically drawn from Schema.org. Each **StatisticalVariable** is marked as a **ConstraintNode**, meaning that some properties (those listed using **constraintProperty**) serve in this setting solely to define the statistical variable rather than literally describe a specific person, place or thing. For example, a **StatisticalVariable** *MedianHeightPerson_Female* representing the median height of women, could be written as follows: the population type is **Person**; the measuredProperty **height**; the **statType** **median**; the **gender** **Female**. It is important to note that there are many kinds of scientific quantitative observation which are not fully, perfectly or unambiguously described following this pattern, or with solely Schema.org terminology. The approach taken here is designed to allow partial, incremental or minimal description of **StatisticalVariables**, and the use of detailed sets of entity and property IDs from external repositories. The **measurementMethod**, **unitCode** and **unitText** properties can also be used to clarify the specific nature and notation of an observed measurement.

Property	Expected Type	Description
Properties from StatisticalVariable		
measuredProperty	Property	The measuredProperty of an Observation , typically via its StatisticalVariable . There are various kinds of applicable Property : a schema.org property, a property from other RDF-compatible systems, e.g. W3C RDF Data Cube, Data Commons, Wikidata, or schema.org extensions such as GS1's .
measurementDenominator	StatisticalVariable	Identifies the denominator variable when an observation represents a ratio or percentage.
measurementMethod	DefinedTerm or MeasurementMethodEnum or Text or URL	A subproperty of measurementTechnique that can be used for specifying specific methods, in particular via MeasurementMethodEnum .
measurementQualifier	Enumeration	Provides additional qualification to an observation. For example, a GDP observation measures the Nominal value.
measurementTechnique	DefinedTerm or MeasurementMethodEnum or Text or URL	A technique, method or technology used in an Observation , StatisticalVariable or Dataset (or DataDownload , DataCatalog), corresponding to the method used for measuring the corresponding variable(s) (for datasets, described using variableMeasured ; for Observation , a StatisticalVariable). Often but not necessarily each variableMeasured will have an explicit representation as (or mapping to) an property such as those defined in Schema.org, or other RDF vocabularies and "knowledge graphs". In that case the subproperty of variableMeasured called measuredProperty is applicable.

Property	Expected Type	Description
		The measurementTechnique property helps when extra clarification is needed about how a measuredProperty was measured. This is oriented towards scientific and scholarly dataset publication but may have broader applicability; it is not intended as a full representation of measurement, but can often serve as a high level summary for dataset discovery. For example, if variableMeasured is: molecule concentration, measurementTechnique could be: "mass spectrometry" or "nmr spectroscopy" or "colorimetry" or "immunofluorescence". If the variableMeasured is "depression rating", the measurementTechnique could be "Zung Scale" or "HAM-D" or "Beck Depression Inventory". If there are several variableMeasured properties recorded for some given data object, use a PropertyValue for each variableMeasured and attach the corresponding measurementTechnique. The value can also be from an enumeration, organized as a MeasurementMethodEnumeration.
populationType	Class	Indicates the populationType common to all members of a StatisticalPopulation or all cases within the scope of a StatisticalVariable .
statType	Property or Text or URL	Indicates the kind of statistic represented by a StatisticalVariable , e.g. mean, count etc. The value of statType is a property, either from within Schema.org (e.g. median , marginOfError , maxValue , minValue) or from other compatible (e.g. RDF) systems such as DataCommons.org or Wikidata.org.
Properties from ConstraintNode		

Property	Expected Type	Description
constraintProperty	Property or URL	Indicates a property used as a constraint. For example, in the definition of a StatisticalVariable . The value is a property, either from within Schema.org or from other compatible (e.g. RDF) systems such as DataCommons.org or Wikidata.org.
numConstraints	Integer	Indicates the number of constraints property values defined for a particular ConstraintNode such as StatisticalVariable . This helps applications understand if they have access to a sufficiently complete description of a StatisticalVariable or other construct that is defined using properties on template-style nodes.
Properties from Thing		
additionalType	Text or URL	An additional type for the item, typically used for adding more specific types from external vocabularies in microdata syntax. This is a relationship between something and a class that the thing is in. Typically the value is a URI-identified RDF class, and in this case corresponds to the use of rdf:type in RDF. Text values can be used sparingly, for cases where useful information can be added without their being an appropriate schema to reference. In the case of text values, the class label should follow the schema.org style guide .
alternateName	Text	An alias for the item.
description	Text or TextObject	A description of the item.
disambiguatingDescription	Text	A sub property of description. A short description of the item used to disambiguate from other, similar items. Information from other properties

Property	Expected Type	Description
		(in particular, name) may be necessary for the description to be useful for disambiguation.
identifier	PropertyValue or Text or URL	The identifier property represents any kind of identifier for any kind of Thing , such as ISBNs, GTIN codes, UUIDs etc. Schema.org provides dedicated properties for representing many of these, either as textual strings or as URL (URI) links. See background notes for more details.
image	ImageObject or URL	An image of the item. This can be a URL or a fully described ImageObject .
mainEntityOfPage	CreativeWork or URL	Indicates a page (or other CreativeWork) for which this thing is the main entity being described. See background notes for details. Inverse property: mainEntity
name	Text	The name of the item.
potentialAction	Action	Indicates a potential Action, which describes an idealized action in which this thing would play an 'object' role.
sameAs	URL	URL of a reference Web page that unambiguously indicates the item's identity. E.g. the URL of the item's Wikipedia page, Wikidata entry, or official website.
subjectOf	CreativeWork or Event	A CreativeWork or Event about this Thing. Inverse property: about
url	URL	URL of the item.

Instances of **StatisticalVariable** may appear as a value for the following properties

Property	On Types	Description
measurementDenominator	Observation or StatisticalVariable	Identifies the denominator variable when an observation represents a ratio or percentage.
variableMeasured	Dataset or Observation	The variableMeasured property can indicate (repeated as necessary) the variables that are measured in some dataset, either described as text or as pairs of identifier and description using PropertyValue, or more explicitly as a StatisticalVariable .

Source

<https://github.com/schemaorg/schemaorg/issues/2564>

Examples

Example 1

No Markup

JSON-LD

Structure

Example encoded as JSON-LD in a HTML script tag.

```
<script type="application/ld+json">
[
{
"@context": "https://schema.org/",
"@type": "StatisticalVariable",
"@id": "Median_Height_Person_Female",
```

```
"name": "Median height of women",
"populationType": {"@id": "Person"},
"measuredProperty": {"@id": "height"},
"statType": {"@id": "median"},
"gender": {"@id": "Female"},
"numConstraints": 1,
"constrainingProperty": {"@id": "gender"}
},
{
"@context": "https://schema.org/",
"@id": "Observation_Median_Age_Person_Female_SanAntonio_TX_2014",
"@type": "Observation",
"name": "Median height of women in San Antonio, Texas in 2014",
"description": "An Observation of the StatisticalVariable Median_Height_Person_Female in location: San Ar
"variableMeasured": { "@id": "Median_Height_Person_Female" },
"observationAbout": { "@id": "https://www.wikidata.org/entity/Q975" },
"observationDate": 2014,
"value": 160,
"unitCode": "CMT"
}
]
</script>
```

